

Machine Learning

CMPSCI 589

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Linear Regression *[Implementation Demo]*

Multivariate Linear Regression

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College of Information
& Computer Sciences



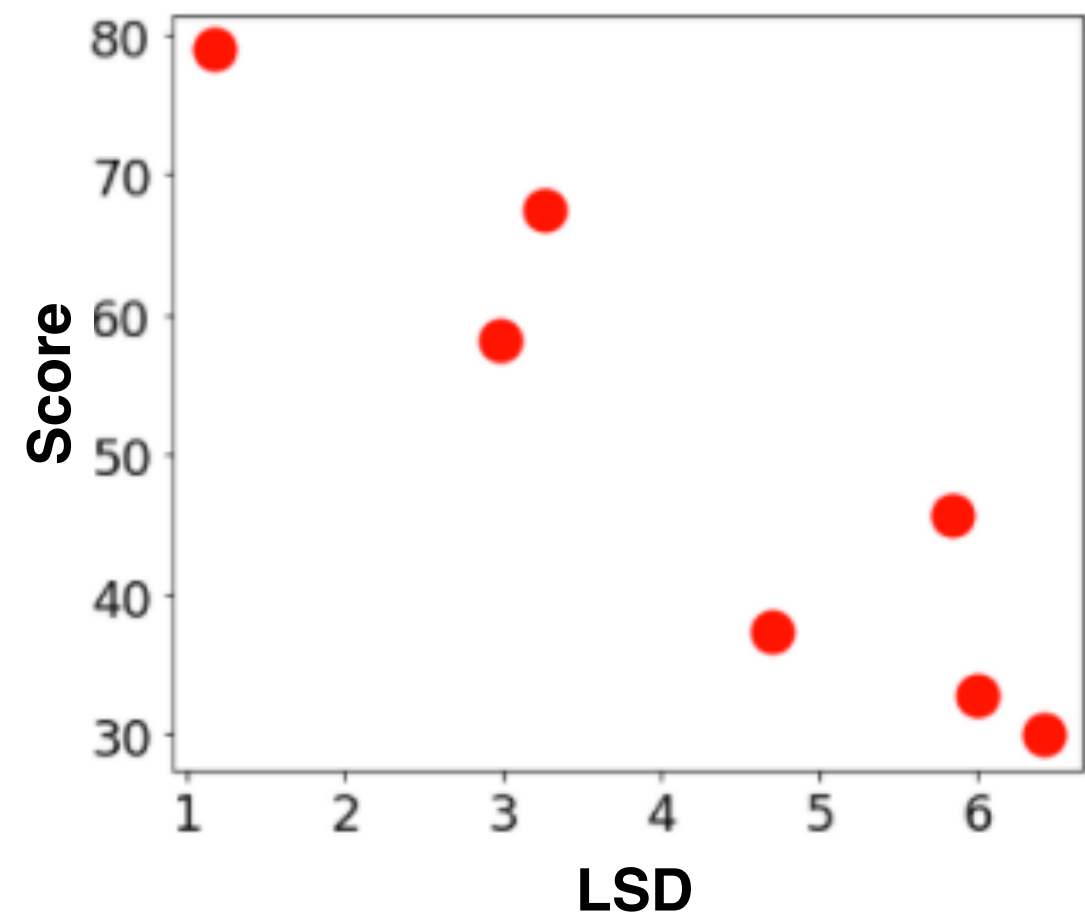
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Review: Regression Problems

- Regression algorithms: predict a real number (i.e., models a function)

	x =LSD Dosage	y = Exam Score
Student1	1.17	78.93
Student2	2.97	58.20
Student3	3.26	67.47
Student4	4.69	37.47
Student5	5.83	45.65
Student6	6.00	32.92
Student7	6.41	29.97



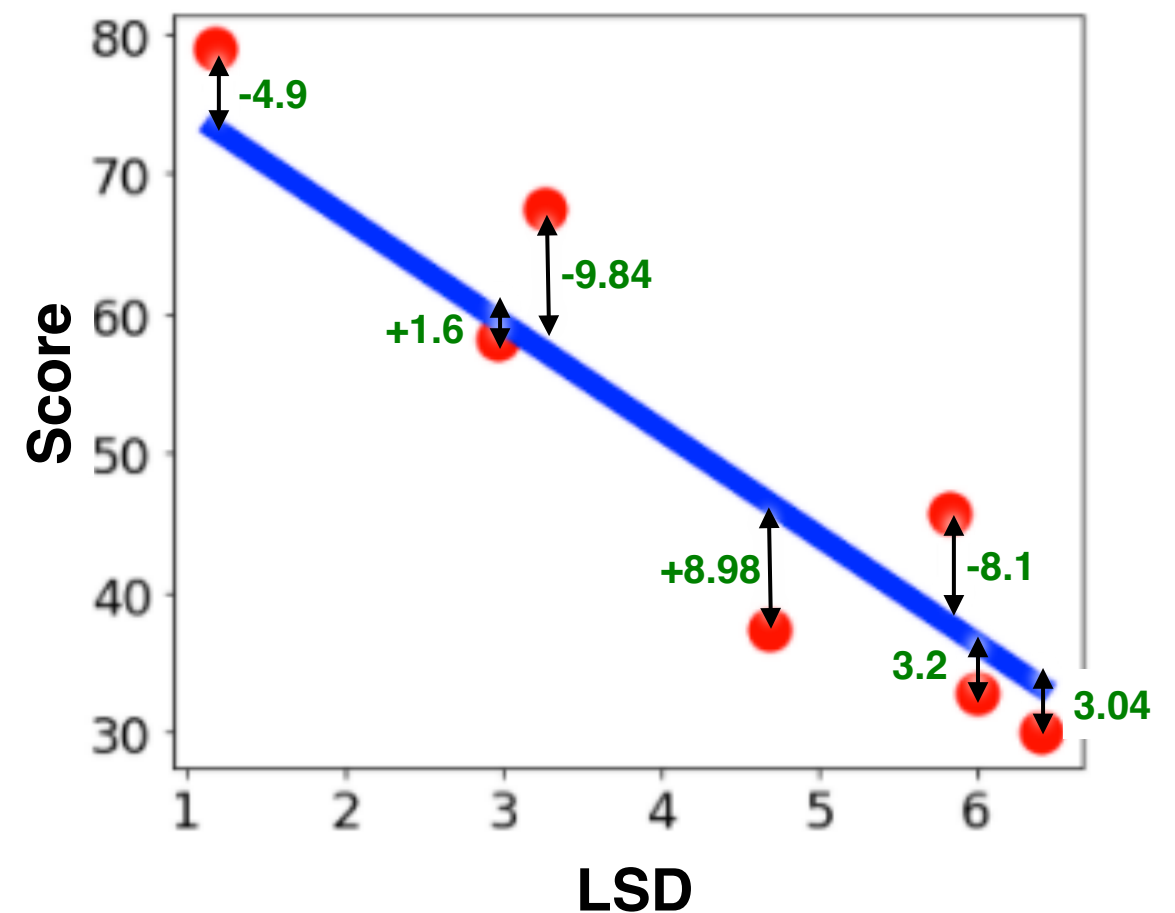
Hypothesis:
 $f(x) = \theta_0 + \theta_1 x$

Assume $\theta_0 = 83.089$
 $\theta_1 = -7.812$

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	x =LSD Dosage	y = Exam Score	f(x) = Predicted Score
Student1	1.17	78.93	73.94
Student2	2.97	58.20	59.88
Student3	3.26	67.47	57.62
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Student5	5.83	45.65	37.54
Student6	6.00	32.92	36.21
Student7	6.41	29.97	33.01



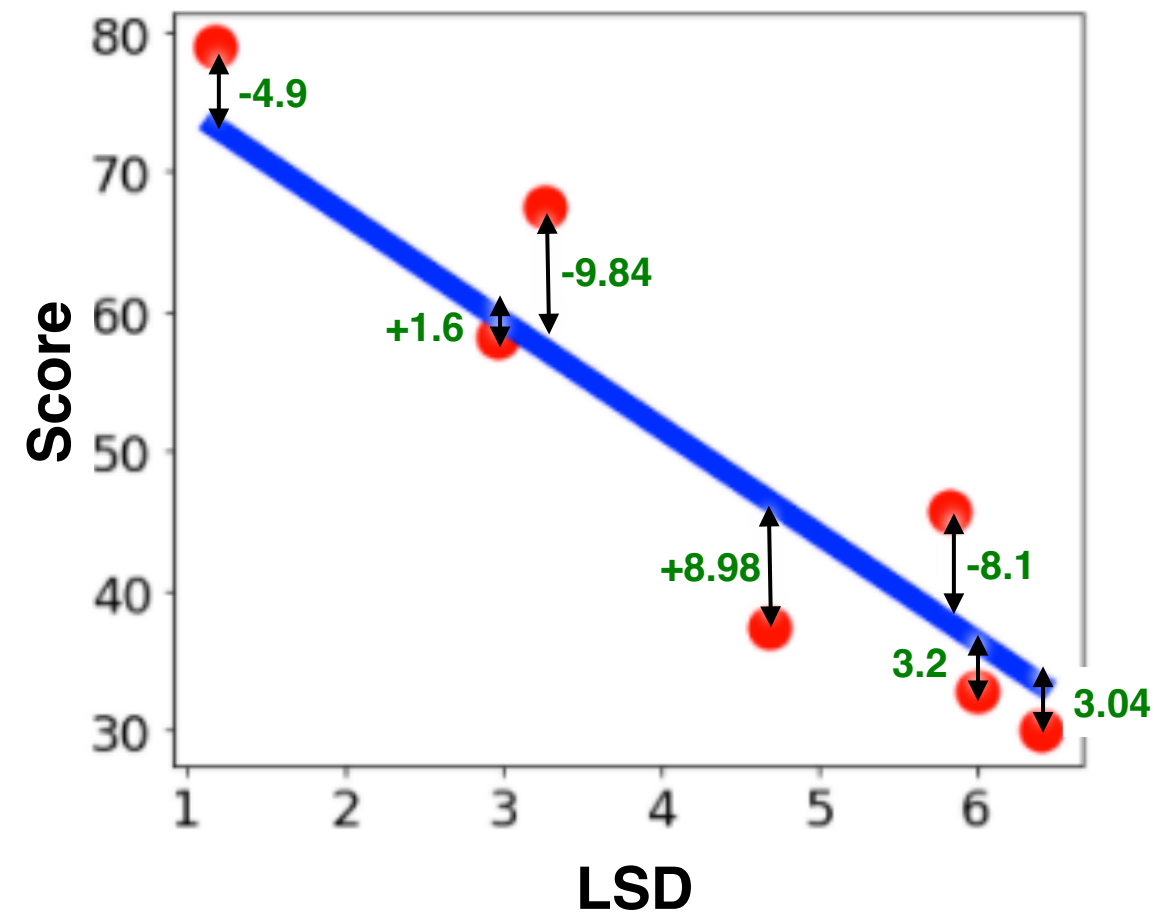
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How "wrong"
is this model?

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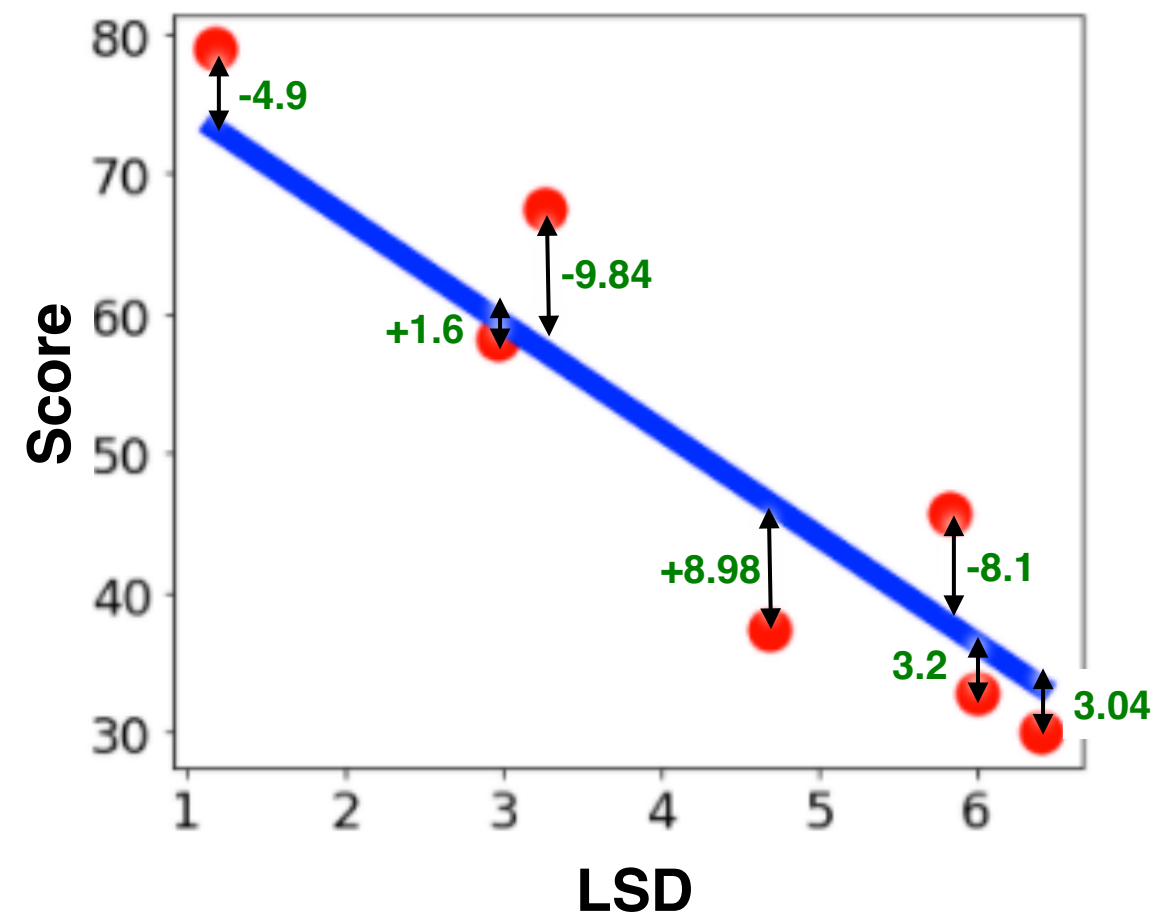
How "wrong" is this model?

Mean squared error:

$$J(\theta_0, \theta_1) = (1/7) \times [(-4.9)^2 + (1.6)^2 + (-9.84)^2 + (8.98)^2 + (-8.1)^2 + (3.2)^2 + (3.04)^2] = 41.30$$

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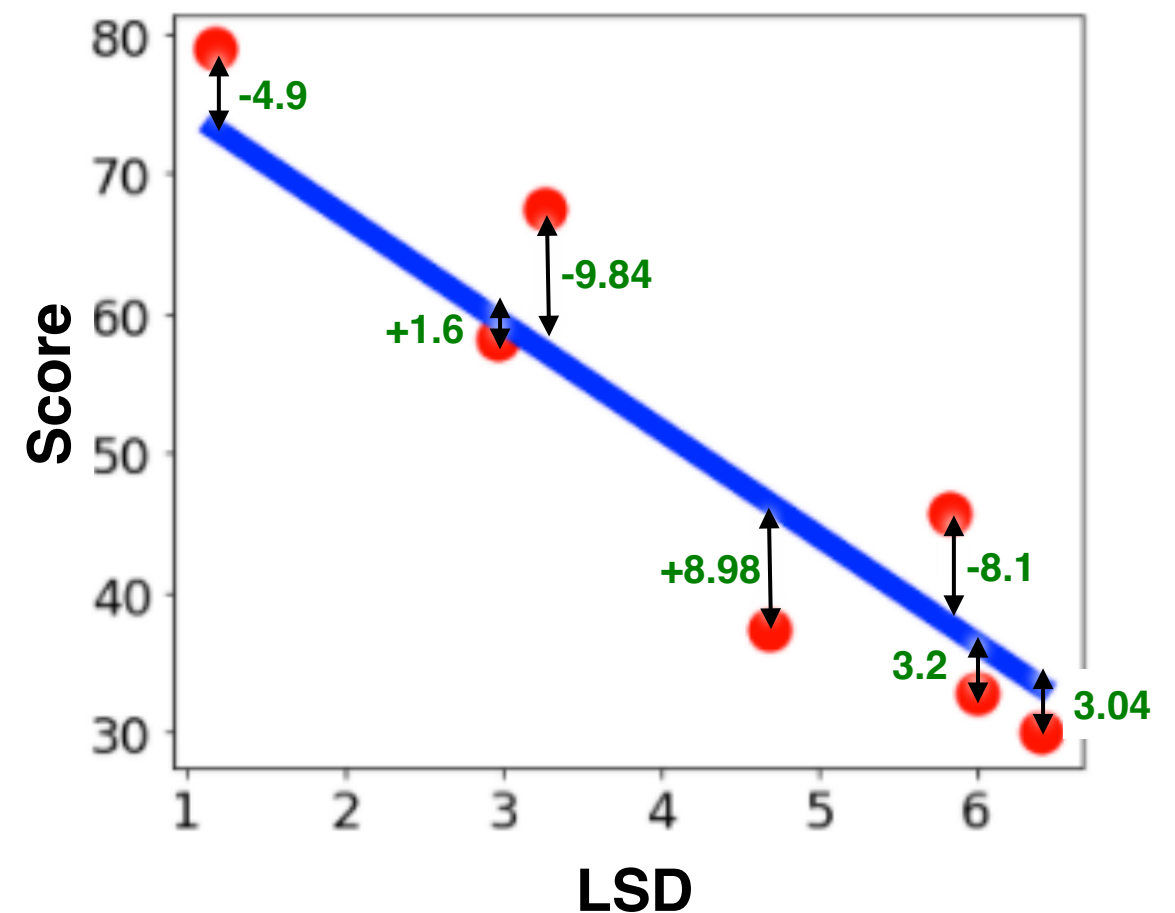
Mean squared error:

$$J(\theta_0, \theta_1) = \frac{1}{n} \times \left[(-4.9)^2 + (1.6)^2 + (-9.84)^2 + (8.98)^2 + (-8.1)^2 + (3.2)^2 + (3.04)^2 \right] = 41.30$$

$$= \sum_{i=1}^n (f_{\theta}(x^{(i)}) - y^{(i)})^2$$

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How "wrong" is this model?

$$J(\theta_0, \theta_1) = \frac{1}{n} \times \left[\sum_{i=1}^n (f_{\theta}(x^{(i)}) - y^{(i)})^2 \right]$$

predicted output given input $x^{(i)}$ correct output given input $x^{(i)}$

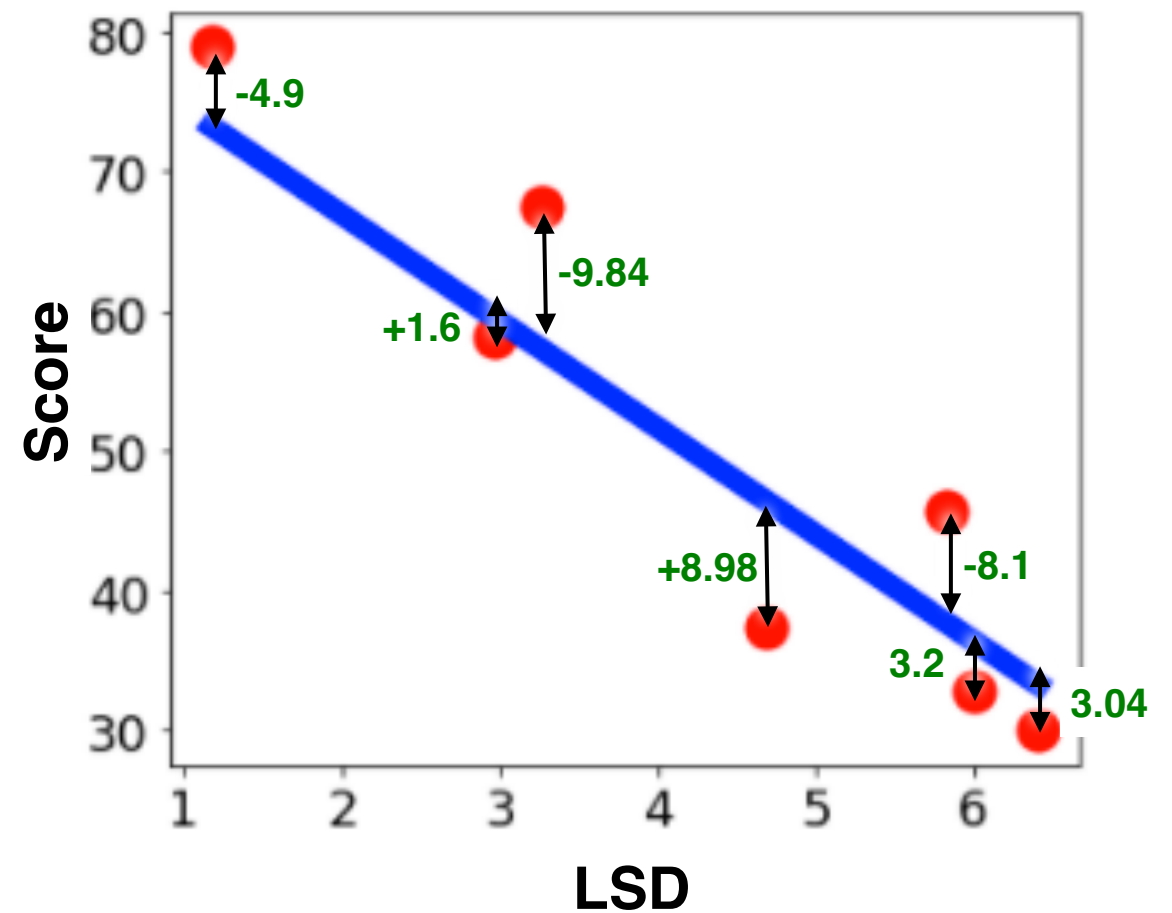
where $f_{\theta}(x) = \theta_0 + \theta_1 x$ is our hypothesis

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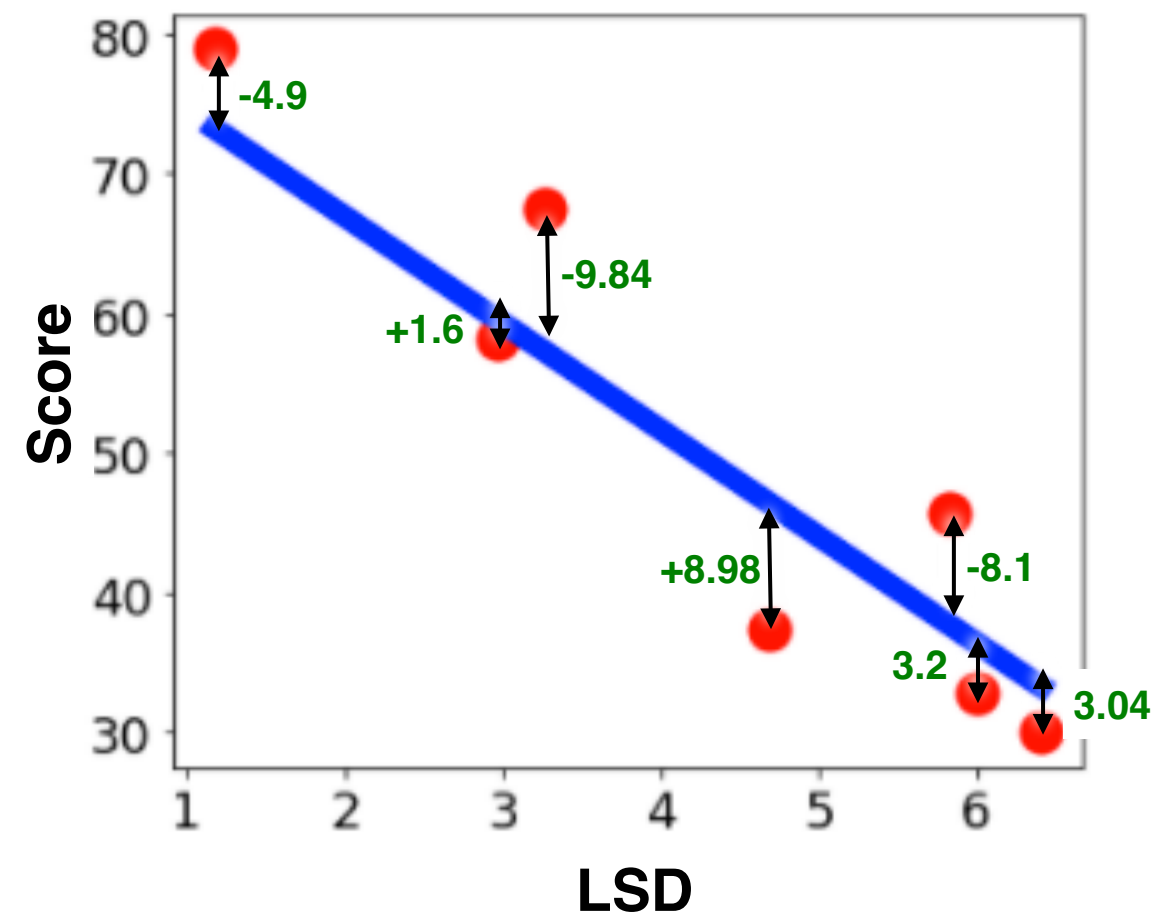


$$J(\theta_0, \theta_1) = \frac{1}{n} \sum_{i=1}^n (f_{\theta}(x^{(i)}) - y^{(i)})^2$$

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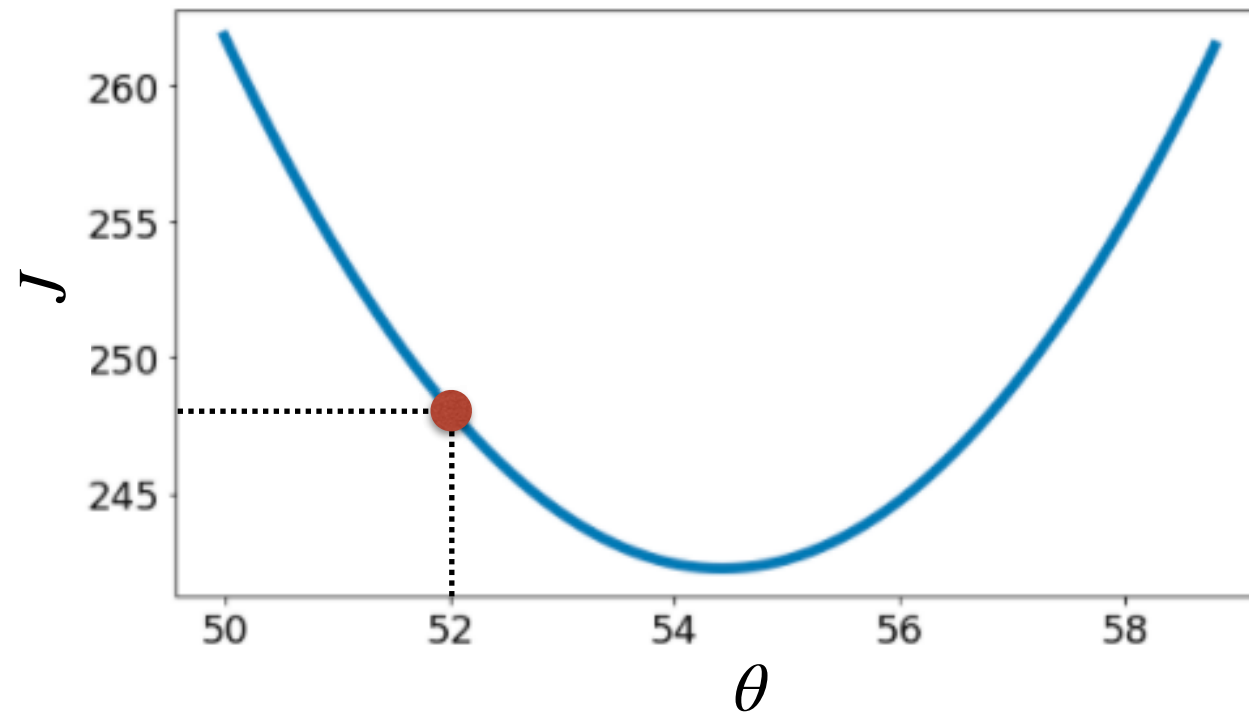
How “wrong”
 is this model?

$$J(\theta_0, \theta_1) = \frac{1}{2n} \sum_{i=1}^n (f_{\theta}(x^{(i)}) - y^{(i)})^2$$

where $f_{\theta}(x) = \theta_0 + \theta_1 x$ is our hypothesis

We wish to find values for θ_0 and θ_1 that minimize the Mean Squared Error, $J(\theta_0, \theta_1)$

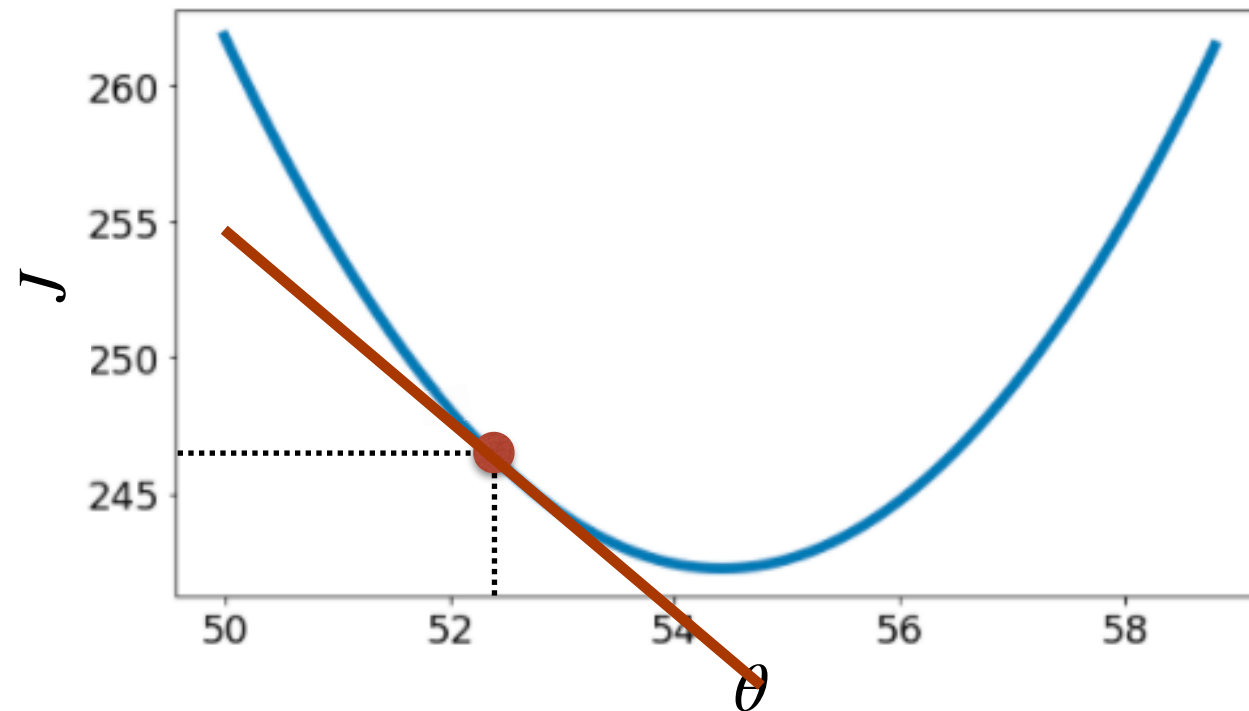
Review: Gradient Descent



How should we update
 θ_0 to decrease the error J ?

Make θ_0 more positive

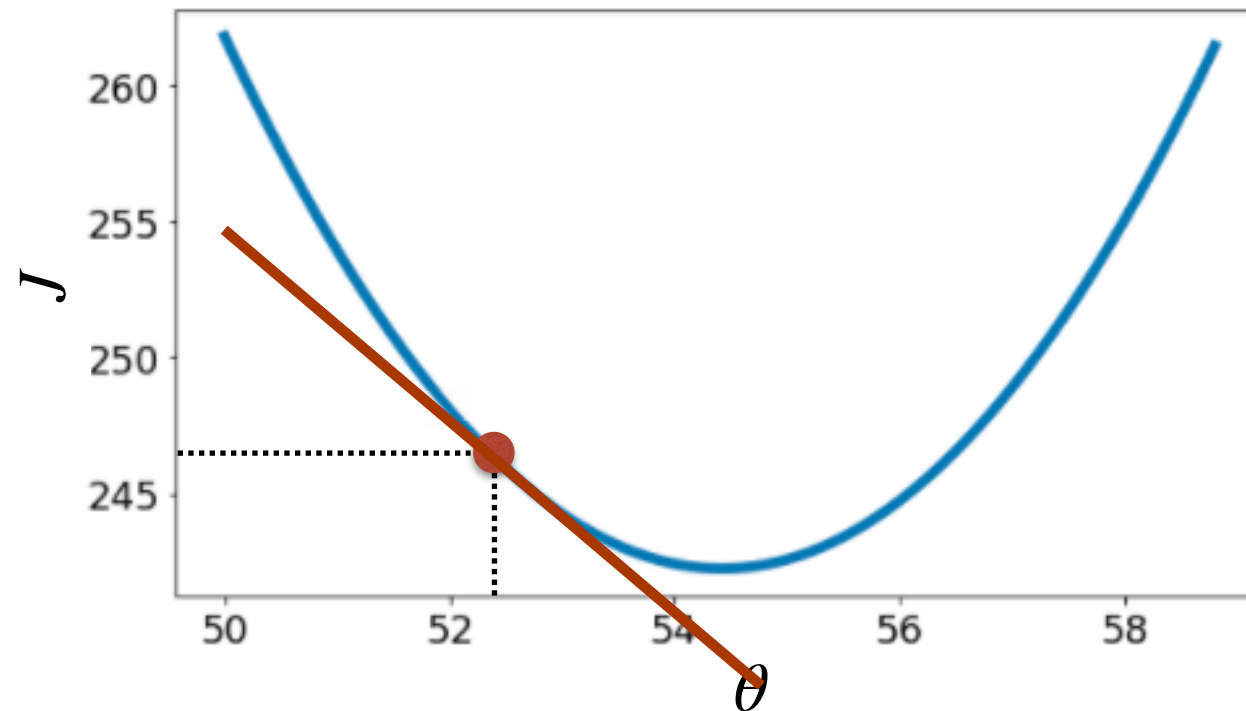
Review: Gradient Descent



Make θ_0 more positive

Derivative of the J function with respect to θ_0
Slope of the function J at that point

Review: Gradient Descent

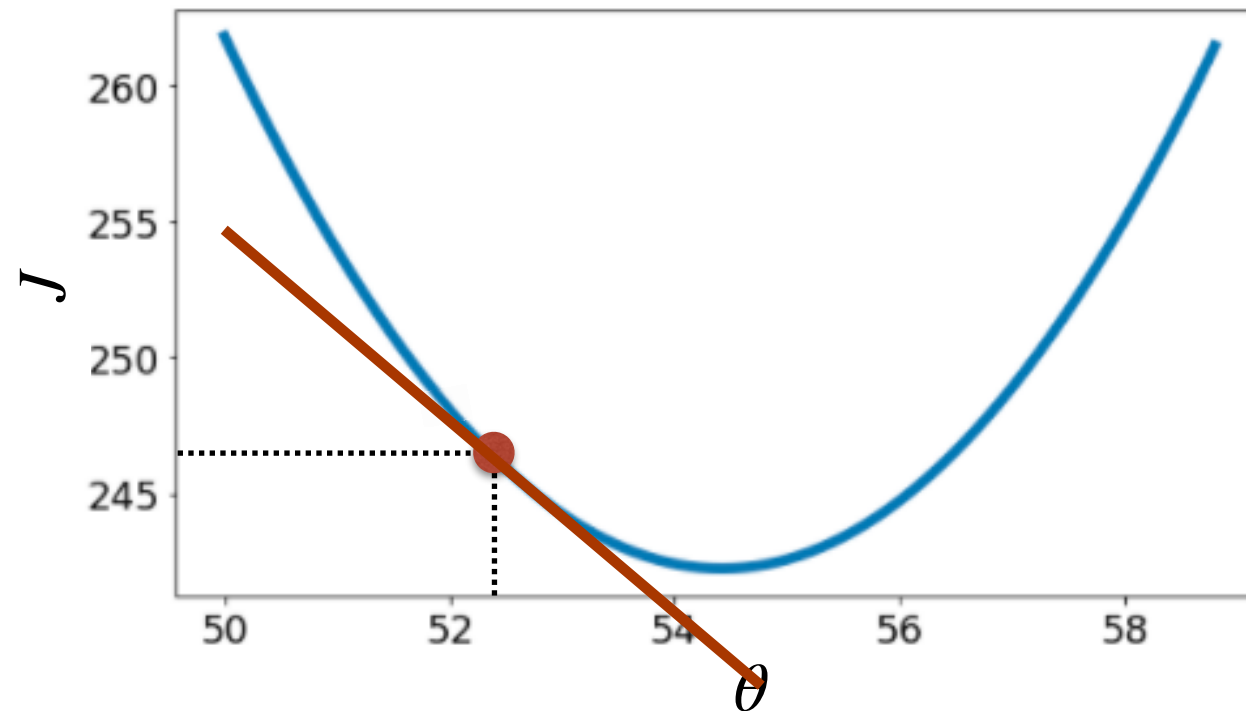


Make θ_0 more positive

Derivative of the J function with respect to θ_0
= -15.22 (negative derivative/slope)

→ if derivative is negative, we should make θ more positive

Review: Gradient Descent



- if derivative is negative, we should make θ more positive
- if derivative is positive, we should make θ more negative

→ To decrease the error J , we should move each θ in the opposite direction of the derivative

Gradient Descent Algorithm

Review: Gradient Descent

We wish to find values for θ_0 and θ_1 that minimize the Mean Squared Error:

$$J(\theta_0, \theta_1) = \frac{1}{2n} \sum_{i=1}^n (f_{\theta}(x^{(i)}) - y^{(i)})^2 \quad \text{where } f_{\theta}(x) = \theta_0 + \theta_1 x \quad \text{is our hypothesis}$$

To decrease the error J , we should move each θ in the opposite direction of the derivative

$$\frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1)$$

$$\theta_0 := \theta_0 - \alpha \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1)$$

$$\frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1)$$

$$\theta_1 := \theta_1 - \alpha \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1)$$

Review: Gradient Descent

We wish to find values for θ_0 and θ_1 that minimize the Mean Squared Error:

$$J(\theta_0, \theta_1) = \frac{1}{2n} \sum_{i=1}^n (f_{\theta}(x^{(i)}) - y^{(i)})^2 \quad \text{where } f_{\theta}(x) = \theta_0 + \theta_1 x \quad \text{is our hypothesis}$$

To decrease the error J , we should move each θ in the opposite direction of the derivative

$$\frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1) = ?$$

$$\frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1) = ?$$

Review: Gradient Descent

$$\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \frac{\partial}{\partial \theta_j} \left(\theta_0 + \theta_1 x^{(i)} \right)$$

$$\frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \times ?$$

Review: Gradient Descent

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Review: Gradient Descent

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Review: Gradient Descent

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Review: Gradient Descent

$$\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \frac{\partial}{\partial \theta_j} \left(\theta_0 + \theta_1 x^{(i)} \right)$$

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Review: Gradient Descent

$$\frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) 1$$
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$$\theta_0 := \theta_0 - \alpha \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1)$$

$$\theta_1 := \theta_1 - \alpha \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1)$$

Implementation

Hypothesis:

$$f_{\theta}(x) = \theta_0 + \theta_1 x$$

**Error function
to be minimized:**

$$J(\theta_0, \theta_1) = \frac{1}{2n} \sum_{i=1}^n (f_{\theta}(x^{(i)}) - y^{(i)})^2$$

**Gradient descent
weight update rule**

$$\theta_0 := \theta_0 - \alpha \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) 1$$

$$\theta_1 := \theta_1 - \alpha \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) x^{(i)}$$

Implementation

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weight update rule

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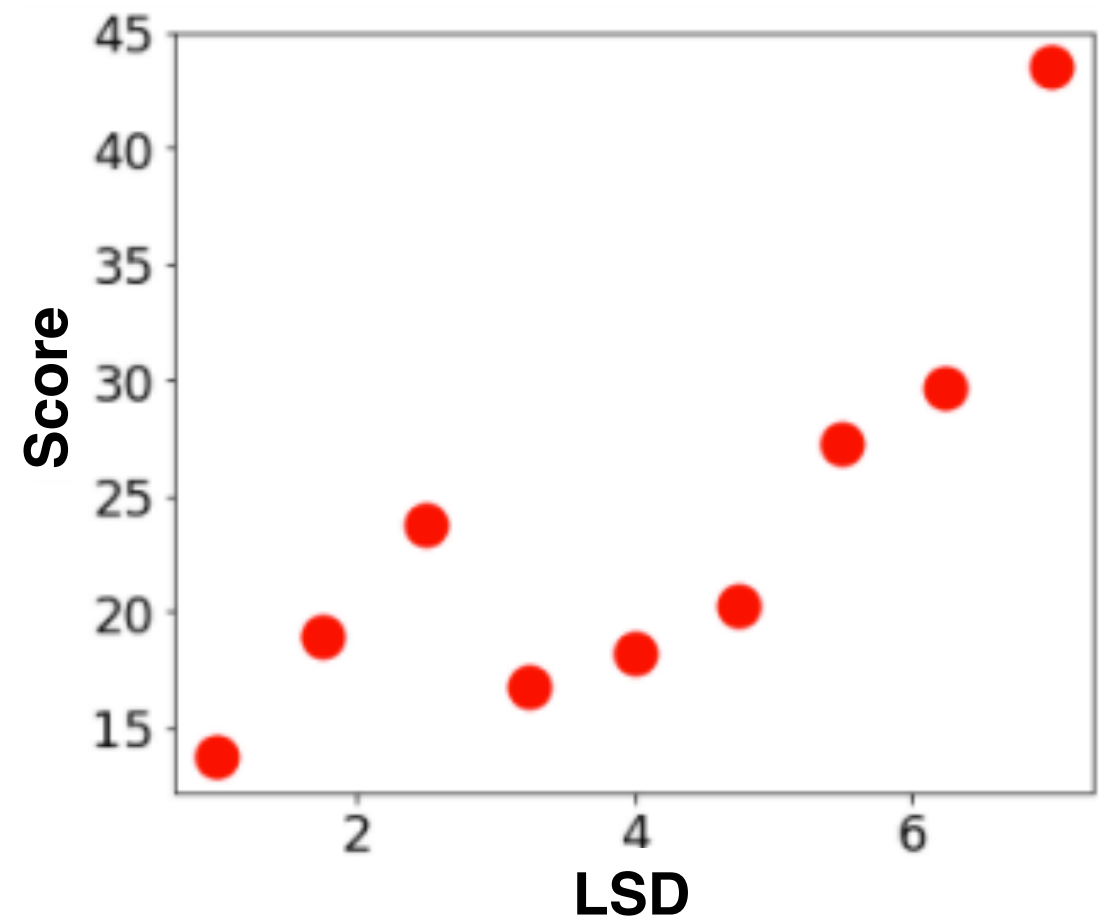
$$\theta_1 := \theta_1 - \alpha \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) x^{(i)}$$

DEMO

Regression Problems

What if the data cannot be approximated well by a straight line?

	x =LSD Dosage	y = Exam Score
Student1	1.0	13.69
Student2	1.75	18.98
Student3	2.5	23.79
Student4	3.25	16.80
Student5	4.0	18.15
Student6	4.75	20.22
Student7	5.5	27.25
Student8	6.25	29.64
Student9	7.0	43.47



We can make a more complex hypothesis

Hypothesis:
 $f(x) = \theta_0 + \theta_1 x + \theta_2 \sin(x)$

Hypothesis: $f_{\theta}(x) = \theta_0 + \theta_1 x + \theta_2 \sin(x)$

$$\begin{aligned}\frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1, \theta_2) &= \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) 1 \\ \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1, \theta_2) &= \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) x^{(i)} \\ \frac{\partial}{\partial \theta_2} J(\theta_0, \theta_1, \theta_2) &= \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \times ?\end{aligned}$$

} like
before

Hypothesis: $f_{\theta}(x) = \theta_0 + \theta_1 x + \theta_2 \sin(x)$

$$\begin{aligned}\frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1, \theta_2) &= \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) 1 \\ \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1, \theta_2) &= \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) x^{(i)} \\ \frac{\partial}{\partial \theta_2} J(\theta_0, \theta_1, \theta_2) &= \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \sin(x^{(i)})\end{aligned} \quad \left. \vphantom{\begin{aligned} \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1, \theta_2) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) 1 \\ \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1, \theta_2) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) x^{(i)} \end{aligned}} \right\} \text{like before}$$

What is the pattern in the update rules?

Hypothesis: $f_{\theta}(x) = \theta_0 + \theta_1 x + \theta_2 \sin(x)$

$$\begin{aligned} \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1, \theta_2) &= \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) 1 \\ \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1, \theta_2) &= \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) x^{(i)} \\ \frac{\partial}{\partial \theta_2} J(\theta_0, \theta_1, \theta_2) &= \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \sin(x^{(i)}) \end{aligned} \quad \left. \vphantom{\begin{aligned} \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1, \theta_2) \\ \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1, \theta_2) \\ \frac{\partial}{\partial \theta_2} J(\theta_0, \theta_1, \theta_2) \end{aligned}} \right\} \text{like before}$$

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DEMO

What is the pattern in the update rules?

$$f_{\theta}(x) = \theta_0 + \theta_1 x + \theta_2 \sin(x)$$

Hypothesis: $f_{\theta}(x) = \theta_1 \phi_1(x) + \theta_2 \phi_2(x) + \dots + \theta_K \phi_K(x)$

$$= \sum_{j=1}^K \theta_j \phi_j(x)$$

where

$$\begin{aligned} \phi_1(x) &= 1 \\ \phi_2(x) &= x \\ \phi_3(x) &= \sin(x) \end{aligned}$$

features

$$\begin{aligned} \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1, \theta_2) &= \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) 1 \\ \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1, \theta_2) &= \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) x^{(i)} \\ \frac{\partial}{\partial \theta_2} J(\theta_0, \theta_1, \theta_2) &= \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \sin(x^{(i)}) \end{aligned} \quad \left. \vphantom{\begin{aligned} \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1, \theta_2) \\ \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1, \theta_2) \\ \frac{\partial}{\partial \theta_2} J(\theta_0, \theta_1, \theta_2) \end{aligned}} \right\} \text{like before}$$

$$f_{\theta}(x) = \theta_0 + \theta_1 x + \theta_2 \sin(x)$$

Hypothesis: $f_{\theta}(x) = \theta_1 \phi_1(x) + \theta_2 \phi_2(x) + \dots + \theta_K \phi_K(x)$

$$= \sum_{j=1}^K \theta_j \phi_j(x) \quad \text{where} \quad \begin{aligned} \phi_1(x) &= 1 \\ \phi_2(x) &= x \\ \phi_3(x) &= \sin(x) \end{aligned}$$

$$\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1, \theta_2) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \frac{\partial}{\partial \theta_j} \left(\quad \quad \quad ? \quad \quad \quad \right)$$

$$f_{\theta}(x) = \theta_0 + \theta_1 x + \theta_2 \sin(x)$$

Hypothesis: $f_{\theta}(x) = \theta_1 \phi_1(x) + \theta_2 \phi_2(x) + \dots + \theta_K \phi_K(x)$

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$$\begin{aligned} \phi_1(x) &= 1 \\ \phi_2(x) &= x \\ \phi_3(x) &= \sin(x) \end{aligned}$$

$$\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1, \theta_2) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \frac{\partial}{\partial \theta_j} \left(\sum_{j=1}^K \theta_j \phi_j(x^{(i)}) \right)$$

$$: \quad \frac{\partial}{\partial \theta_j} J(\theta_1, \dots, \theta_K) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \phi_j(x^{(i)})$$

$$f_{\theta}(x) = \theta_0 + \theta_1 x + \theta_2 \sin(x)$$

Hypothesis: $f_{\theta}(x) = \theta_1 \phi_1(x) + \theta_2 \phi_2(x) + \dots + \theta_K \phi_K(x)$

$$= \sum_{j=1}^K \theta_j \phi_j(x)$$

where

$$\begin{aligned} \phi_1(x) &= 1 \\ \phi_2(x) &= x \\ \phi_3(x) &= \sin(x) \end{aligned}$$

$$\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1, \theta_2) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \frac{\partial}{\partial \theta_j} \left(\sum_{j=1}^K \theta_j \phi_j(x^{(i)}) \right)$$

$$: \frac{\partial}{\partial \theta_j} J(\theta_1, \dots, \theta_K) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \phi_j(x^{(i)})$$

Linear Regression: General Formulation

Hypothesis:

$$\begin{aligned} f_{\theta}(x) &= \theta_1 \phi_1(x) + \theta_2 \phi_2(x) + \dots + \theta_K \phi_K(x) \\ &= \sum_{j=1}^K \theta_j \phi_j(x) \end{aligned}$$

**Error function
to be minimized:**

$$J(\theta_1, \dots, \theta_K) = \frac{1}{2n} \sum_{i=1}^n (f_{\theta}(x^{(i)}) - y^{(i)})^2$$

**Gradient descent
weight update rule**

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta_1, \dots, \theta_K)$$

where

$$\frac{\partial}{\partial \theta_j} J(\theta_1, \dots, \theta_K) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \phi_j(x^{(i)})$$

Linear Regression - Implementation

From $t=1 \dots \infty$

1. For $j=1$ to K // computes gradients

1.1 Compute
$$\frac{\partial}{\partial \theta_j} J(\theta_1, \dots, \theta_K) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \phi_j(x^{(i)})$$

2. For $j=1$ to K // updates weights

2.1
$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta_1, \dots, \theta_K)$$

3. If $|J_t(\theta_1, \dots, \theta_K) - J_{t-1}(\theta_1, \dots, \theta_K)| < \varepsilon$ // stopping criterion

3.1 Return $\theta_1, \dots, \theta_K$

Multivariate Regression

What if there are many input variables?

	$x_1 =$ Area	$x_2 =$ #Bedrooms	$x_3 =$ #Floors	$x_4 =$ How_Old	$y =$ Price
House1	2100	5	1	45	460
House2	1410	3	2	40	232
House3	1534	3	2	30	315
House4	850	2	1	36	180
...	...				

$x_1^{(1)}$
 $x_1^{(2)}$
 $x_4^{(3)}$

(source: Machine Learning course, Andrew Ng, Coursera)

$f(\text{Area, \#Bedrooms, \#Floors, How_Old}) \rightarrow \text{Price}$

$f(x_1, x_2, x_3, x_4) \rightarrow y$

$x^{(i)}$ features of the i -th house $\longrightarrow$ e.g., $x^{(2)} = [1410, 3, 2, 40]$
 $x_j^{(i)}$ j -th feature of the i -th house $\longrightarrow$ e.g., $x_4^{(2)} = 40$

Multivariate Regression

	$x_1 =$ Area	$x_2 =$ #Bedrooms	$x_3 =$ #Floors	$x_4 =$ How_Old	$y =$ Price
House1	2100	5	1	45	460
House2	1410	3	2	40	232
House3	1534	3	2	30	315
House4	850	2	1	36	180
...	...				

$f(\text{Area}, \text{\#Bedrooms}, \text{\#Floors}, \text{How_Old}) \rightarrow \text{Price}$

$f(x_1, x_2, x_3, x_4) \rightarrow y$

$x^{(i)}$ features of the i-th house

$x_j^{(i)}$ j-th feature of the i-th house

(source: Machine Learning course, Andrew Ng, Coursera)

A simple hypothesis

$$f_{\theta}(x_1, x_2, x_3, x_4) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \theta_4 x_4$$

where

$$\phi_1(x) = 1 \quad \phi_2(x) = x_1 \quad \phi_3(x) = x_2 \quad \phi_4(x) = x_3 \quad \phi_5(x) = x_4$$

Multivariate Regression

	$x_1 =$ Area	$x_2 =$ #Bedrooms	$x_3 =$ #Floors	$x_4 =$ How_Old	$y =$ Price
House1	2100	5	1	45	460
House2	1410	3	2	40	232
House3	1534	3	2	30	315
House4	850	2	1	36	180
...	...				

$f(\text{Area, \#Bedrooms, \#Floors, How_Old}) \rightarrow \text{Price}$

$f(x_1, x_2, x_3, x_4) \rightarrow y$

$x^{(i)}$ features of the i-th house

$x_j^{(i)}$ j-th feature of the i-th house

(source: Machine Learning course, Andrew Ng, Coursera)

A more complex hypothesis

$$f_{\theta}(x_1, x_2, x_3, x_4) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \theta_4 x_4 + \theta_5 \left(\frac{x_1}{x_2} \right) + \theta_6 \left(\frac{x_2}{x_3} \right) + \theta_7 \left(\sqrt{x_1} \right) = \sum_{j=1}^K \theta_j \phi_j(x)$$

where

$$\begin{aligned} \phi_1(x) &= 1 & \phi_2(x) &= x_1 & \phi_3(x) &= x_2 & \phi_4(x) &= x_3 & \phi_5(x) &= x_4 \\ \phi_6(x) &= \left(\frac{x_1}{x_2} \right) & \phi_7(x) &= \left(\frac{x_2}{x_3} \right) & \phi_8(x) &= \left(\sqrt{x_1} \right) \end{aligned}$$

Linear Regression: General Formulation

Hypothesis:

$$\begin{aligned} f_{\theta}(x) &= \theta_1 \phi_1(x) + \theta_2 \phi_2(x) + \dots + \theta_K \phi_K(x) \\ &= \sum_{j=1}^K \theta_j \phi_j(x) \end{aligned}$$

**Error function
to be minimized:**

$$J(\theta_1, \dots, \theta_K) = \frac{1}{2n} \sum_{i=1}^n (f_{\theta}(x^{(i)}) - y^{(i)})^2$$

**Gradient descent
weight update rule**

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta_1, \dots, \theta_K)$$

where

$$\frac{\partial}{\partial \theta_j} J(\theta_1, \dots, \theta_K) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \phi_j(x^{(i)})$$

Feature Normalization

	$x_1 =$ Area	$x_2 =$ #Bedrooms	$x_3 =$ #Floors	$x_4 =$ How_Old	$y =$ Price
House1	2100	5	1	45	460
House2	1410	3	2	40	232
House3	1534	3	2	30	315
House4	850	2	1	36	180
...	...				

Problem: range of values of each feature are very different

- normalize/transform features so they have the same range/interval
 - (e.g., $[0,1]$ ou $[-1,1]$)

Feature Normalization



	$x_1 =$ Area	$x_2 =$ #Bedrooms	$x_3 =$ #Floors	$x_4 =$ How_Old	$y =$ Price
House1	1.0	5	1	45	460
House2	0.448	3	2	40	232
House3	0.5472	3	2	30	315
House4	0.0	2	1	36	180
...	...				

Problem: range of values of each feature are very different

- normalize/transform features so they have the same range/interval
 - (e.g., $[0,1]$ ou $[-1,1]$)

Feature Normalization



	$x_1 =$ Area	$x_2 =$ #Bedrooms	$x_3 =$ #Floors	$x_4 =$ How_Old	$y =$ Price
House1	1.0	1.0	1	45	460
House2	0.448	0.33	2	40	232
House3	0.5472	0.33	2	30	315
House4	0.0	0	1	36	180
...	...				

Problem: range of values of each feature are very different

- normalize/transform features so they have the same range/interval
 - (e.g., $[0,1]$ ou $[-1,1]$)

Feature Normalization



	$x_1 =$ Area	$x_2 =$ #Bedrooms	$x_3 =$ #Floors	$x_4 =$ How_Old	$y =$ Price
House1	1.0	1.0	0.0	45	460
House2	0.448	0.33	1.0	40	232
House3	0.5472	0.33	1.0	30	315
House4	0.0	0	0.0	36	180
...	...				

Problem: range of values of each feature are very different

- normalize/transform features so they have the same range/interval
 - (e.g., $[0,1]$ ou $[-1,1]$)

Feature Normalization



	$x_1 =$ Area	$x_2 =$ #Bedrooms	$x_3 =$ #Floors	$x_4 =$ How_Old	$y =$ Price
House1	1.0	1.0	0.0	1.0	460
House2	0.448	0.33	1.0	0.66	232
House3	0.5472	0.33	1.0	0.0	315
House4	0.0	0	0.0	0.4	180
...	...				

Problem: range of values of each feature are very different

- normalize/transform features so they have the same range/interval
 - (e.g., $[0,1]$ ou $[-1,1]$)

Feature Normalization



	$x_1 =$ Area	$x_2 =$ #Bedrooms	$x_3 =$ #Floors	$x_4 =$ How_Old	$y =$ Price
House1	1.0	1.0	0.0	1.0	1.0
House2	0.448	0.33	1.0	0.66	0.18
House3	0.5472	0.33	1.0	0.0	0.48
House4	0.0	0	0.0	0.4	0.0
...	...				

Problem: range of values of each feature are very different

- normalize/transform features so they have the same range/interval
 - (e.g., $[0,1]$ ou $[-1,1]$)

Feature Normalization



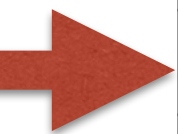
	$x_1 =$ Area	$x_2 =$ #Bedrooms	$x_3 =$ #Floors	$x_4 =$ How_Old	$y =$ Price
House1	1.0	1.0	0.0	1.0	1.0
House2	0.448	0.33	1.0	0.66	0.18
House3	0.5472	0.33	1.0	0.0	0.48
House4	0.0	0	0.0	0.4	0.0
...	...				

Problem: range of values of each feature are very different

- normalize/transform features so they have the same range/interval
 - (e.g., $[0,1]$ ou $[-1,1]$)

WHY?

Feature Normalization



	$x_1 =$ Area	$x_2 =$ #Bedrooms	$x_3 =$ #Floors	$x_4 =$ How_Old	$y =$ Price
House1	2100	5	1	45	460
House2	1410	3	2	40	232
House3	1534	3	2	30	315
House4	850	2	1	36	180
...	...				

Intuition:

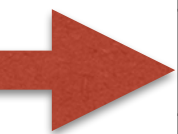
- values of x_1 between 0-2100
- values of x_2 between 1-5

$$f_{\theta}(x_1, x_2, x_3, x_4) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \theta_4 x_4$$

Assume we are using the following weights to predict the price of House1

$$f_{\theta}(x_1, x_2, x_3, x_4) = 0.0 + 0.15x_1 + 0.5x_2 + 0.3x_3 + 4.0x_4 = 497.8$$

Feature Normalization



	$x_1 =$ Area	$x_2 =$ #Bedrooms	$x_3 =$ #Floors	$x_4 =$ How_Old	$y =$ Price
House1	2100	5	1	45	460
House2	1410	3	2	40	232
House3	1534	3	2	30	315
House4	850	2	1	36	180
...	...				

Intuition:

- values of x_1 between 0-2100
- values of x_2 between 1-5

Assume we are using the following weights to predict the price of House1

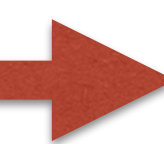
$$f_{\theta}(x_1, x_2, x_3, x_4) = 0.0 + 0.15x_1 + 0.5x_2 + 0.3x_3 + 4.0x_4 = 497.8$$

What is the impact of increasing weight of x_1 (that is, θ_1) by +0.1?

$$f_{\theta}(x_1, x_2, x_3, x_4) = 0.0 + 0.25x_1 + 0.5x_2 + 0.3x_3 + 4.0x_4 = 707.8$$

Weight changed a little (+0.1)
Significant change in prediction
(from 497.8 to 707.8)

Feature Normalization



	$x_1 =$ Area	$x_2 =$ #Bedrooms	$x_3 =$ #Floors	$x_4 =$ How_Old	$y =$ Price
House1	2100	5	1	45	460
House2	1410	3	2	40	232
House3	1534	3	2	30	315
House4	850	2	1	36	180
...	...				

Intuition:

- values of x_1 between 0-2100
- values of x_2 between 1-5

Assume we are using the following weights to predict the price of House1

$$f_{\theta}(x_1, x_2, x_3, x_4) = 0.0 + 0.15x_1 + 0.5x_2 + 0.3x_3 + 4.0x_4 = 497.8$$

What is the impact of increasing weight of x_1 (that is, θ_1) by +0.1?

$$f_{\theta}(x_1, x_2, x_3, x_4) = 0.0 + 0.25x_1 + 0.5x_2 + 0.3x_3 + 4.0x_4 = 707.8$$

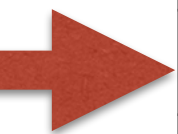
Weight changed a little (+0.1)
Significant change in prediction
(from 497.8 to 707.8)

What is the impact of increasing weight of x_2 (that is, θ_2) by +0.1?

$$f_{\theta}(x_1, x_2, x_3, x_4) = 0.0 + 0.15x_1 + 0.6x_2 + 0.3x_3 + 4.0x_4 = 498.3$$

Weight changed a little (+0.1)
Small change in prediction
(from 497.8 to 498.3)

Feature Normalization



	$x_1 =$ Area	$x_2 =$ #Bedrooms	$x_3 =$ #Floors	$x_4 =$ How_Old	$y =$ Price
House1	2100	5	1	45	460
House2	1410	3	2	40	232
House3	1534	3	2	30	315
House4	850	2	1	36	180
...	...				

Intuition:

- values of x_1 between 0-2100
- values of x_2 between 1-5

WHY?

Assume we are using the following weights to predict the price of House1

$$f_{\theta}(x_1, x_2, x_3, x_4) = 0.0 + 0.15x_1 + 0.5x_2 + 0.3x_3 + 4.0x_4 = 497.8$$

What is the impact of increasing weight of x_1 (that is, θ_1) by +0.1?

$$f_{\theta}(x_1, x_2, x_3, x_4) = 0.0 + 0.25x_1 + 0.5x_2 + 0.3x_3 + 4.0x_4 = 707.8$$

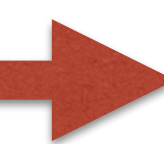
Weight changed a little (+0.1)
Significant change in prediction
(from 497.8 to 707.8)

What is the impact of increasing weight of x_2 (that is, θ_2) by +0.1?

$$f_{\theta}(x_1, x_2, x_3, x_4) = 0.0 + 0.15x_1 + 0.6x_2 + 0.3x_3 + 4.0x_4 = 498.3$$

Weight changed a little (+0.1)
Small change in prediction
(from 497.8 to 498.3)

Feature Normalization



	$x_1 =$ Area	$x_2 =$ #Bedrooms	$x_3 =$ #Floors	$x_4 =$ How_Old	$y =$ Price
House1	2100	5	1	45	460
House2	1410	3	2	40	232
House3	1534	3	2	30	315
House4	850	2	1	36	180
...	...				

Intuition:

- values of x_1 between 0-2100
- values of x_2 between 1-5

values of x_1 are significantly larger x_2

- changing its weight drastically changes the prediction f_θ

Assume we are using the following weights to predict the price of House1

$$f_\theta(x_1, x_2, x_3, x_4) = 0.0 + 0.15x_1 + 0.5x_2 + 0.3x_3 + 4.0x_4 = 497.8$$

What is the impact of increasing weight of x_1 (that is, θ_1) by +0.1?

$$f_\theta(x_1, x_2, x_3, x_4) = 0.0 + 0.25x_1 + 0.5x_2 + 0.3x_3 + 4.0x_4 = 707.8$$

Weight changed a little (+0.1)
Significant change in prediction
(from 497.8 to 707.8)

What is the impact of increasing weight of x_2 (that is, θ_2) by +0.1?

$$f_\theta(x_1, x_2, x_3, x_4) = 0.0 + 0.15x_1 + 0.6x_2 + 0.3x_3 + 4.0x_4 = 498.3$$

Weight changed a little (+0.1)
Small change in prediction
(from 497.8 to 498.3)

Feature Normalization

	x₁ = Area	x₂ = #Bedrooms	x₃ = #Floors	x₄ = How_Old	y = Price
House1	2100	5	1	45	460
House2	1410	3	2	40	232
House3	1534	3	2	30	315
House4	850	2	1	36	180
...	...				

Intuition:

- values of **x₁** between **0-2100**
- values of **x₂** between **1-5**

values of x₁ are significantly larger x₂

- changing its weight drastically changes the prediction **f_θ**

Feature normalization

- transform all attributes/features to a same range
- typically **[0, 1]** or **[-1, +1]**

$$\max_1 = 2100$$

$$\min_1 = 850$$

$$x_2^1 = \frac{1410 - 850}{2100 - 850} = 0.448$$

$$\max_i := \max(x_i^1, \dots, x_i^n)$$

$$\min_i := \min(x_i^1, \dots, x_i^n)$$

$$x_i^d := \frac{x_i^d - \min_d}{\max_d - \min_d}$$

**Normalizing
to [0, 1]**

Feature Normalization

	$x_1 =$ Area	$x_2 =$ #Bedrooms	$x_3 =$ #Floors	$x_4 =$ How_Old	$y =$ Price
House1	2100	5	1	45	460
House2	0.448	3	2	40	232
House3	1534	3	2	30	315
House4	850	2	1	36	180
...	...				

Intuition:

- values of x_1 between 0-2100
- values of x_2 between 1-5

values of x_1 are significantly larger x_2

- changing its weight drastically changes the prediction f_θ

Feature normalization

- transform all attributes/features to a same range
- typically **[0, 1]** or **[-1, +1]**

$$\max_1 = 2100$$

$$\min_1 = 850$$

$$x_2^1 = \frac{1410 - 850}{2100 - 850} = 0.448$$

$$\max_i := \max(x_i^1, \dots, x_i^n)$$

$$\min_i := \min(x_i^1, \dots, x_i^n)$$

$$x_i^d := \frac{x_i^d - \min_d}{\max_d - \min_d}$$

**Normalizing
to [0, 1]**

Feature Normalization

	x₁ = Area	x₂ = #Bedrooms	x₃ = #Floors	x₄ = How_Old	y = Price
House1	2100	5	1	45	460
House2	1410	3	2	40	232
House3	1534	3	2	30	315
House4	850	2	1	36	180
...	...				

Intuition:

- values of **x₁** between **0-2100**
- values of **x₂** between **1-5**

values of x₁ are significantly larger x₂

- changing its weight drastically changes the prediction **f_θ**

Feature normalization

- transform all attributes/features to a same range
- typically **[0, 1]** or **[-1, +1]**

$$\max_1 = 2100$$

$$\min_1 = 850$$

$$x_2^1 = 2 \left(\frac{1410 - 850}{2100 - 850} \right) - 1 = -0.104$$

$$\max_i := \max(x_i^1, \dots, x_i^n)$$

$$\min_i := \min(x_i^1, \dots, x_i^n)$$

$$x_i^d := 2 \left(\frac{x_i^d - \min_d}{\max_d - \min_d} \right) - 1$$

**Normalizing
to [-1, 1]**

Feature Normalization

	$x_1 =$ Area	$x_2 =$ #Bedrooms	$x_3 =$ #Floors	$x_4 =$ How_Old	$y =$ Price
House1	2100	5	1	45	460
House2	-0.104	3	2	40	232
House3	1534	3	2	30	315
House4	850	2	1	36	180
...	...				

Intuition:

- values of x_1 between 0-2100
- values of x_2 between 1-5

values of x_1 are significantly larger x_2

- changing its weight drastically changes the prediction f_θ

Feature normalization

- transform all attributes/features to a same range
- typically **[0, 1]** or **[-1, +1]**

$$\max_1 = 2100$$

$$\min_1 = 850$$

$$x_2^1 = 2 \left(\frac{1410 - 850}{2100 - 850} \right) - 1 = -0.104$$

$$\max_i := \max(x_i^1, \dots, x_i^n)$$

$$\min_i := \min(x_i^1, \dots, x_i^n)$$

$$x_i^d := 2 \left(\frac{x_i^d - \min_d}{\max_d - \min_d} \right) - 1$$

**Normalizing
to [-1, 1]**

Linear Regression - Implementation

Normalize all input variables, $\mathbf{x}$, and output variables, $\mathbf{y}$

From $t=1 \dots \infty$

1. For $j=1$ to K // computes gradients

1.1 Compute
$$\frac{\partial}{\partial \theta_j} J(\theta_1, \dots, \theta_K) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) \phi_j(x^{(i)})$$

2. For $j=1$ to K // updates weights

2.1
$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta_1, \dots, \theta_K)$$

3. If $|J_t(\theta_1, \dots, \theta_K) - J_{t-1}(\theta_1, \dots, \theta_K)| < \varepsilon$ // stopping criterion

3.1 Return $\theta_1, \dots, \theta_K$



Machine Learning

CMPSCI 589

Bruno C. da Silva

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Linear Regression *[Implementation Demo]*

Multivariate Linear Regression

UMassAmherst

College of Information
& Computer Sciences



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